# HOMOLOGY ISOMORPHISM FROM A LEVEL-N SHEAF-SURJECTIVE COSKELETON MAP

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 879-893 of sources/stacks-hypercovering/pinned-original-source.tex; proof span is lines 895-932, the complete author proof (36 body lines).

*Editorial note.* Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 13 of queue_hypercovering.json (pb=1406, stmt 879-893, proof 895-932).

*Editorial note.* stacks-hypercovering wave-1 (hc01): fresh adjudication accept (2026-10-14); pool per standing fallback order (hypercovering -> pic -> spaces-more-cohomology) after md01 closed the moduli pool; hypercovering.tex 53 labels/34 with proof judged 26 accept / 8 reject / 14 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 01GC: sources/stacks-hypercovering/pinned-original-source.tex, source lines 808–824. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 01A6: sources/stacks-hypercovering/pinned-original-source.tex, source lines 932–932. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 08NS: sources/stacks-hypercovering/pinned-original-source.tex, source lines 932–932. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $\mathcal{C}$ be a site. Let $f : L \to K$ be a morphism of simplicial objects of $PSh(\mathcal{C})$. Let $n \geq 0$ be an integer. Assume that*

(1) *For $i < n$ the morphism $L_i \to K_i$ is an isomorphism.*
(2) *The morphism $L_n \to K_n$ is surjective after sheafification.*
(3) *The canonical map $L \to cosk_n sk_n L$ is an isomorphism.*

*Date*: stacks-project:hypercovering.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(4) *The canonical map $K \to cosk_n sk_n K$ is an isomorphism.*

*Then $H_i(f) : H_i(L) \to H_i(K)$ is an isomorphism.*

*Proof.* This proof is exactly the same as the proof of Lemma 01GC above. Namely, we first let $K'_n \subset K_n$ be the sub presheaf which is the image of the map $L_n \to K_n$. Assumption (2) means that the sheafification of $K'_n$ is equal to the sheafification of $K_n$. Moreover, since $L_i = K_i$ for all $i < n$ we see that get an $n$-truncated simplicial presheaf $U$ by taking $U_0 = L_0 = K_0, \ldots, U_{n-1} = L_{n-1} = K_{n-1}, U_n = K'_n$. Denote $K' = \text{cosk}_n U$, a simplicial presheaf. Because we can construct $K'_m$ as a finite limit, and since sheafification is exact, we see that $(K'_m)^\# = K_m$. In other words, $(K')^\# = K^\#$. We conclude, by exactness of sheafification once more, that $H_i(K) = H_i(K')$. Thus it suffices to prove the lemma for the morphism $L \to K'$, in other words, we may assume that $L_n \to K_n$ is a surjective morphism of *presheaves*!

In this case, for any object $U$ of $\mathcal{C}$ we see that the morphism of simplicial sets

$$L(U) \longrightarrow K(U)$$

satisfies all the assumptions of Simplicial, Lemma 01A6. Hence it is a trivial Kan fibration. In particular it is a homotopy equivalence (Simplicial, Lemma 08NS). Thus

$$\mathbf{Z}_L(U) \longrightarrow \mathbf{Z}_K(U)$$

is a homotopy equivalence too. This for all $U$. The result follows. □